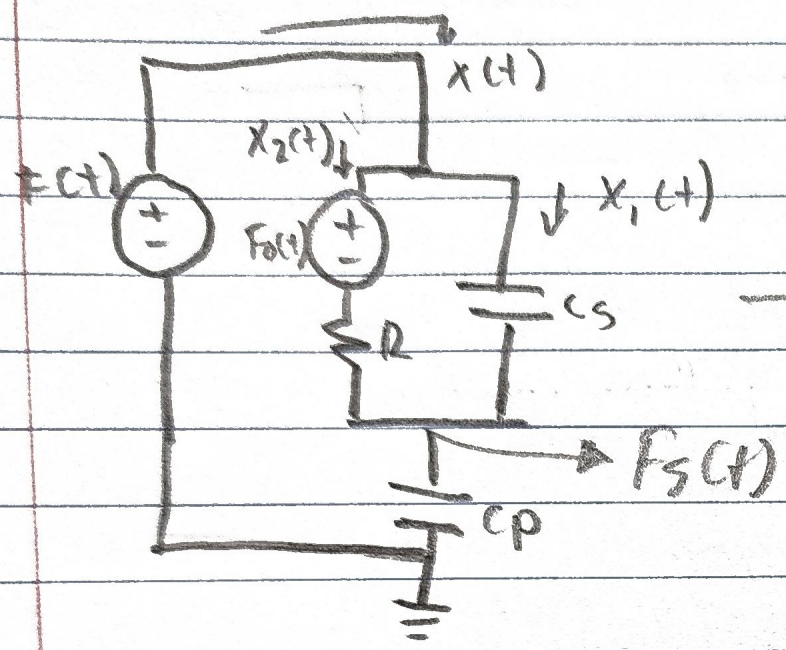
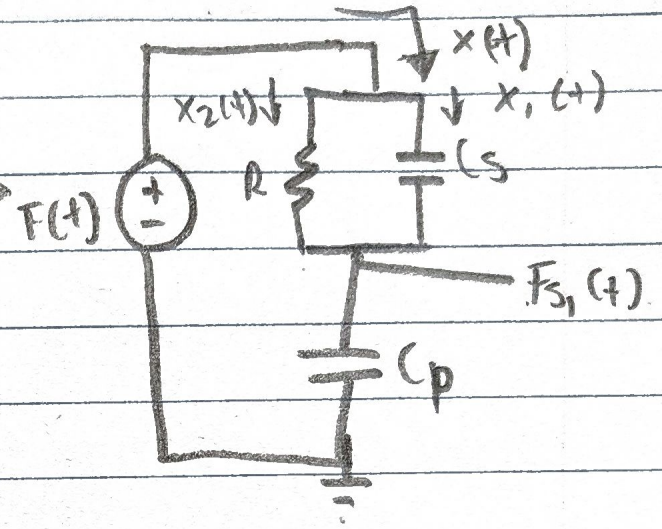
28-Oct-26

Modelado y control del sistema musculoesquelético

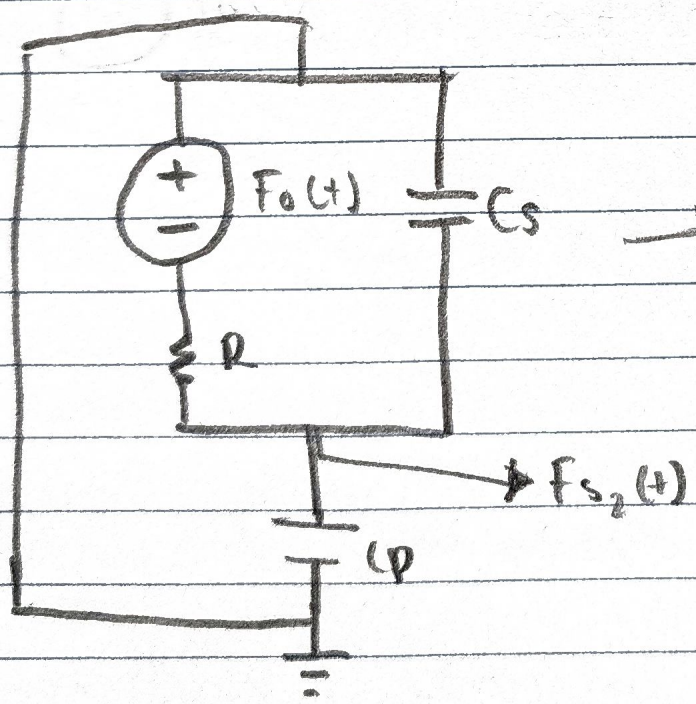
Circuito eléctrico

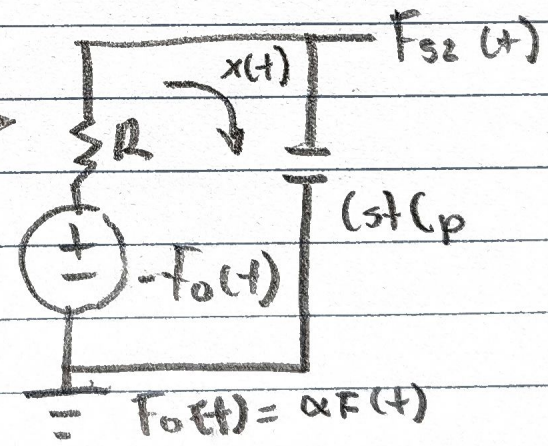


Apagando $F_0(t)$ C(1)

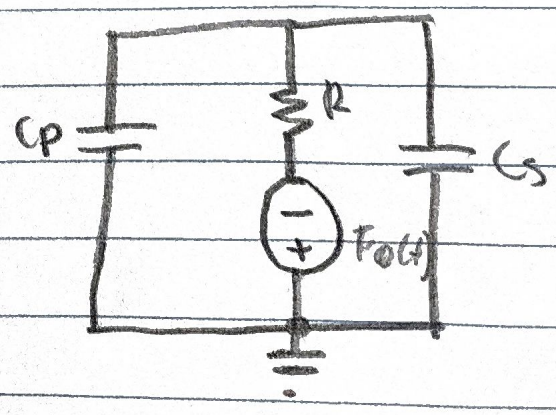


Apagando F(t) C(2)





$$F_S(t) = F_{S_1}(t) + F_{S_2}(t)$$



Análisis del circuito por superposición

Apagando $F_o(t)$ Circuito (1)

$$X(t) = X_1(t) + X_2(t) \qquad X_1(t) = C_s \frac{d[F(t) - F_{s1}(t)]}{dt}$$

$$X(t) = C_p \frac{dF_{s1}(t)}{dt} \qquad X_2(t) = \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + \frac{F(s) - F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

$$\left(C_p s + C_s s + \frac{1}{R}\right) F_{s1}(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

$$F_{s1}(s) = \frac{\frac{C_s R s + 1}{R} F(s)}{\frac{R(C_p + C_s)s + 1}{R}} = \frac{C_s R s + 1}{R(C_p + C_s)s + 1} F(s)$$

Apagando $F(t)$

$$-\alpha F(t) = R x(t) + \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$F_{s2}(t) = \frac{1}{C_p + C_s} \int x(t)\, dt$$

$$-\alpha F(s) = R X(s) + \frac{X(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{X(s)}{(C_p + C_s)s}$$

$$F(s) = -\frac{X(s)}{\alpha} \frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}$$

$$\frac{F_{S2}(s)}{F(s)} = \frac{-\frac{X(s)}{\alpha}}{\frac{\frac{X(s)}{C(R+Cs)s}}{\frac{R(Cs+CR)s+1}{(R+Cs)s}}} = \frac{\alpha}{R(CR+Cs)s+1}$$

$$F_{S2}(s) = -\frac{\alpha F(s)}{R(CR+Cs)s+1}$$

Función de transferencia

$$F_S(s) = F_{S1}(s) + F_{S2}(s) = \frac{CsRs+1}{R(CR+Cs)s+1} F(s) - \frac{\alpha F(s)}{R(CR+Cs)s+1}$$

$$F_S(s) = \frac{CsRs+1-\alpha}{R(CR+Cs)s+1} F(s)$$

$$\frac{F_S(s)}{F(s)} = \frac{CsRs+1-\alpha}{R(CR+Cs)s+1}$$